		Target engagement	Dose	Efficacy	Safety
Autoimmune	**psoriasis** anti-IL23A antibodies (3)	**2023** Acute local intradermal human IL-23 mouse skin assay dosed with mirikizumab	**2018** Cynomolgus monkey repeat-dose toxicology (NOAEL)	**2019** Human psoriasis skin xenograft on AGR129 mice (anti-p19 clone 7G10)	**2018** Mouse intracellular-infection challenge with surrogate anti-p19 (BCG, Salmonella, Listeria)
	ulcerative colitis S1PR1 agonists (3)	**2016** Rodent blood lymphocyte counts (ozanimod)	**2023** Rat and dog repeat-dose toxicology (etrasimod)	**2016** TNBS and CD4+CD45RBhi transfer colitis mice (ozanimod)	**2023** hERG current assay in vitro (etrasimod) △ Ozanimod still slows the heart
Endocrine	**endometriosis** GNRHR antagonists (3)	**2022** Rat HPG axis (LH in male rats; E2 in pro-oestrus female rats), linzagolix	**2022** Linzagolix PK-PD systems-pharmacology model	**2024** Linzagolix PK-PD systems-pharmacology model	**2022** Linzagolix PK-PD systems-pharmacology model △ Tox blind to BMD loss; liver held
	secondary hyperparathyroidism CASR positive modulators (4)	**2013** 5/6-nephrectomy uremic rat	**2017** Population PK/PD model of etelcalcetide (PTH-calcium homeostasis, CaSR cooperativity)	**2015** 5/6-nephrectomy uremic rat	**2023** Rat gastric emptying and marmoset emesis vs cinacalcet
	type 1 diabetes mellitus SLC5A2 inhibitors (4)	**2014** STZ-diabetic rat	**2008** STZ-diabetic rat	**2018** STZ-diabetic rat	**2018** Diet-induced obese rat, insulin-replete, dapagliflozin, 38 days △ DKA 3-4x placebo, missed in rats
GI	**GERD** ATP4A blockers (3)	**2019** Stimulated gastric acid secretion in dogs	**2017** Stimulated gastric acid secretion in dogs	**2019** Rat reflux oesophagitis vs esomeprazole	**2025** Rat repeat dosing: gastrin, emptying, microbiota
	primary biliary cholangitis NR1H4 agonists (3)	**2024** Rodent FXR target-gene PD models	**2019** Rodent FXR target-gene PD models	**2021** Mdr2-/- mice	**2020** Mdr2-/- mice Ductular-reaction prediction never tested in PBC
Hematology	**hemophilia A** F8 replacement / gene therapy (8)	**2025** Cynomolgus macaques with pre-existing anti-AAV5 antibodies given valoctocogene roxaparvovec	**2017** F8-knockout mice + cynomolgus macaques	**2021** Hemophilia A dogs (Chapel Hill / Queen's colonies) △ Mouse-stable BAX 888 lost in man	**2023** Hemophilia A dogs (Chapel Hill / Queen's colonies) △ ALT rise in 81% of patients
	hemophilia B F9 replacement / gene therapy (7)	**2017** Hemophilia B dogs, AAV8 canine FIX-Padua △ FIX expression lost in 7 of 8	**2017** Hemophilia B dogs, AAV8 canine FIX-Padua	**2024** Juvenile hemophilia B dogs, canine surrogate vector	**2024** Cynomolgus GLP tox, fidanacogene predecessor
	PNH anti-C5 antibodies (4)	**2022** Hemolysis assay with C5 R885H variant	**2020** Cynomolgus PK/PD (crovalimab)	**2018** Complement hemolysis assay in human serum	**2020** Drug–target–drug complex assay, human C5
Metabolic	**hyperlipidemia** CETP inhibitors (3)	**2014** Human CETP/apoA-I transgenic mouse	**2014** CETP/apoA-I mouse; rat blood pressure △ Exposure far below prediction	**2011** Human CETP loss-of-function genotypes ☐ Outcome trial negative	**2014** Rat blood pressure + human adrenal H295R cells
	hyperlipidemia HCAR2 agonists (3)	**2015** Rat, guinea pig plasma NEFA	**2017** Rat, guinea pig plasma NEFA Exposure reached, no HDL-C response	**2017** Rat, guinea pig plasma NEFA GSK256073 phase 2a stopped for futility	**2017** Guinea pig ear-temperature flush model (GSK)
Oncology	**myelodysplastic syndrome** HDAC (pan) inhibitors (3)	**2017** Mouse xenografts (HCT-116, later MOLM-13/SET-2 AML)	**2017** Mouse xenografts (HCT-116, later MOLM-13/SET-2 AML) RP2D reached, combination failed	**2017** Mouse AML xenografts (MOLM-13, SET-2) CR 18% vs 33% placebo	**2015** Dog telemetry and rat/dog repeat-dose toxicology (panobinostat IND package)
	small cell lung carcinoma anti-PDCD1 antibodies (3)	**2023** In-vitro PD-1 occupancy of toripalimab on human T cells, saturating cynomolgus tox	**2018** Cynomolgus tox, nivolumab Target reached; CheckMate 331 missed OS	**2021** Tumour-mutational-burden rule, SCLC exomes, nivolumab △ Missed OS, indication withdrawn	**2014** PD-1 knockout mice △ Fatal pneumonitis, not in monkeys
Ophthalmology	**diabetic macular edema** anti-VEGFA antibodies (3)	**2011** STZ-diabetic rat, Evans-blue blood-retinal-barrier leak (intravitreal VEGF Trap, VGT-NC-007)	**2023** Rabbit chronic retinal-neovascularisation leak model (aflibercept 8 mg)	**2021** STZ-diabetic rat, Evans-blue blood-retinal-barrier leak (intravitreal KH902/conbercept)	**2023** Monkey ocular tox, aflibercept PHOTON: no nasomucosal signal at 8 mg
	dry eye syndrome anti-TNF antibodies (3)	**2023** Aged mouse dry eye	**2013** Rabbit topical ocular PK (ESBA105 eye drops)	**2022** Canine dry eye (spontaneous) Dog sign benefit; Phase 3 flat vs placebo	**2022** Rabbit topical ocular PK (systemic exposure margin)
Psy.	**insomnia** HCRTR1/2 antagonists (6)	**2013** Rat, dog and rhesus sleep EEG/EMG (suvorexant)	**2022** PBPK-PD model of brain OX2 receptor occupancy (daridorexant, 65% occupancy threshold)	**2012** Rat, dog and rhesus sleep EEG/EMG (suvorexant)	**2019** Mouse emotion-induced cataplexy (lemborexant) △ Animals no abuse; humans liked it
Respiratory	**cystic fibrosis** CFTR correctors (10)	**2024** FRT cells expressing CFTR variants △ Third corrector added nothing	**2015** Rat and dog toxicology (lumacaftor)	**2024** FRT cells expressing CFTR variants	**2025** Rat and dog tox (vanzacaftor)
	cystic fibrosis CFTR potentiators (5)	**2026** Patient-derived primary nasal epithelial cells (HNE, Ussing chamber theratyping)	**2022** Human hepatic metabolism in vitro, scaled to PK	**2024** FRT cells carrying each CFTR variant (Ussing chamber)	**2024** Rat toxicology (juvenile and adult, ivacaftor program)
Uro.	**interstitial cystitis** anti-NGF antibodies (3)	**1997** Rodent chemical cystitis	**1997** Rodent chemical cystitis	**2022** ASP6294 vs anti-NGF IgG in arthritic rodents ASP6294 missed IC pain endpoint	**2012** Primate and rodent joint tox Joint harm missed; IC trials put on hold
Other	**atopic eczema** HRH4 antagonists (3)	**2014** Mouse FITC dermatitis, H4R	**2021** Mouse FITC dermatitis, H4R 17-fold adriforant dose range, nothing worked	**2021** Dog acute atopic dermatitis, JNJ 7777120 and JNJ 28307474 △ Adriforant: mice yes, patients no	**2014** Rat, monkey 6/9-month tox Agranulocytosis the tox species never showed
	atopic eczema PDE4 inhibitors (5)	**2019** Stimulated human blood cells (PBMC cytokine release)	**2016** Minipig dermal toxicity (3- and 9-month)	**2023** Mouse chronic allergic contact dermatitis (difamilast) △ Explant engaged; primary failed	**2016** Rabbit primary skin irritation
	atopic eczema anti-IL33 antibodies (4)	**2024** IL-33 Tg/KO mice, MC903	**2024** IL-33 Tg/KO mice, MC903 Target saturated at 300 mg; still failed	**2021** Anti-IL-33 in DNCB mice Three more antibodies, three more misses	**2021** Cynomolgus PK and IL-1-family specificity panel (torudokimab)